

Test Case: Study Group

TC-SG-GROUP-001

Title: Create a valid study group

Steps:

1. Log in and navigate to the Study Group Hub.
2. Select **Create New Group**.
3. Enter a valid group name, description, and icon.
4. Click the button to create the group.

Expected Result:

- The study group is created successfully.
- The creator is assigned the **LEADER** role.
- The system automatically generates a group access code.
- The new group appears in the creator's group list.

Priority: High

TC-SG-GROUP-002

Title: Verify that the generated group code is automatic and unique

Steps:

1. Log in with multiple valid user accounts.
2. Create several different study groups.
3. Record the access code generated for each group.
4. Compare the generated group codes.

Expected Result:

- Each group receives a 6-character access code.
- The code contains only uppercase letters and numbers.
- No two groups have the same access code.

Priority: High

TC-SG-GROUP-003

Title: Validate group information length limits

Steps:

1. Log in and open the group creation or group update form.
2. Enter a group name longer than 100 characters.
3. Enter a description longer than 250 characters or an icon value longer than 50 characters.

4. Submit the create or update request.

Expected Result:

- The system rejects input that exceeds the allowed length limits.
- An appropriate validation message is displayed.
- The group information is not created or updated with invalid data.

Priority: Medium

TC-SG-GROUP-004

Title: Join a study group using a nonexistent access code

Steps:

1. Log in and open the **Join Group** function.
2. Enter a nonexistent 6-character group access code.
3. Click **Join Group**.

Expected Result:

- The user is not added to any group.
- The system displays the message **Invalid group access code** or an equivalent error message.

Priority: High

TC-SG-GROUP-005

Title: Prevent a user from joining the same group twice

Steps:

1. Log in with an account that is already a member of a study group.
2. Open the **Join Group** function.
3. Enter the access code of the group the user has already joined.
4. Click **Join Group**.

Expected Result:

- No duplicate group membership record is created.
- The system informs the user that they are already a member of the group.

Priority: High

TC-SG-ROLE-006

Title: Leader updates study group information

Steps:

1. Log in as the Leader of the study group.
2. Open the group editing page.
3. Change the group name, description, or icon.
4. Save the changes.

Expected Result:

- The group information is updated successfully.
- The updated information is displayed correctly in both the Study Group Workspace and Study Group Hub.

Priority: High

TC-SG-ROLE-007

Title: Member is not allowed to update study group information**Steps:**

1. Log in as a Member of the study group.
2. Attempt to access the group update function directly through the interface or API.
3. Submit updated group information.

Expected Result:

- The request is rejected because the Member does not have permission to update the group.
- The study group information remains unchanged.

Priority: High

TC-SG-ROLE-008

Title: Group member views group information and member list**Steps:**

1. Log in as either the Leader or a Member of the study group.
2. Open the Study Group Workspace.
3. View the group information and member list.

Expected Result:

- The correct group name, group code, and member count are displayed.
- Each member's name, avatar, role, and join date are displayed correctly.

Priority: Medium

TC-SG-ROLE-009

Title: Leader cannot remove themselves from the study group

Steps:

1. Log in as the Leader of the study group.
2. Attempt to remove the Leader's own account from the group.
3. Observe the system response.

Expected Result:

- The system rejects the request.
- The Leader remains a member of the study group.
- The system informs the Leader that leadership must be transferred before leaving or removing themselves from the group.

Priority: High

TC-SG-ROLE-010

Title: Leader removes a Member from the study group**Preconditions:**

- The study group has one Leader and at least one Member.

Steps:

1. The Leader selects a Member from the group member list.
2. Select **Remove Member** and confirm the action.
3. The removed Member attempts to access the Study Group Workspace again.

Expected Result:

- The Member is removed from the study group member list.
- The total number of group members decreases by one.
- The removed Member can no longer access the study group's resources or workspace.

Priority: High

Test Execution

Test Case ID	Execution Date	Status	Actual Result	Bug ID
TC-SG-GROUP-001	20/08/2026	Pass	Group was created successfully, the creator was assigned the Leader role, and a unique group code was generated.	N/A
TC-SG-GROUP-002	20/08/2026	Pass	Each created group received a unique 6-character access code containing uppercase letters and numbers.	N/A
TC-SG-GROUP-	20/08/2026	Pass	The system correctly rejected group information that exceeded the allowed length limits and displayed	N/A

Test Case ID	Execution Date	Status	Actual Result	Bug ID
003			validation messages.	
TC-SG-GROUP-004	20/08/2026	Pass	The system rejected the nonexistent group access code and did not add the user to any group.	N/A
TC-SG-GROUP-005	20/08/2026	Pass	The system prevented duplicate membership and informed the user that they were already a member of the group.	N/A
TC-SG-ROLE-006	20/08/2026	Pass	The Leader successfully updated the group information, and the new data was displayed correctly in the workspace and Study Group Hub.	N/A
TC-SG-ROLE-007	20/08/2026	Pass	The Member was denied permission to update group information, and the existing group data remained unchanged.	N/A
TC-SG-ROLE-008	20/08/2026	Pass	The group information and member list were displayed correctly, including member name, avatar, role, and join date.	N/A
TC-SG-ROLE-009	20/08/2026	Pass	The system prevented the Leader from removing themselves from the group and kept the Leader membership unchanged.	N/A
TC-SG-ROLE-010	20/08/2026	Pass	The Leader successfully removed the selected Member, the member count decreased by one, and the removed Member could no longer access group resources.	N/A